

PIXEL Resolution.

THE PICTURE ELEMENT

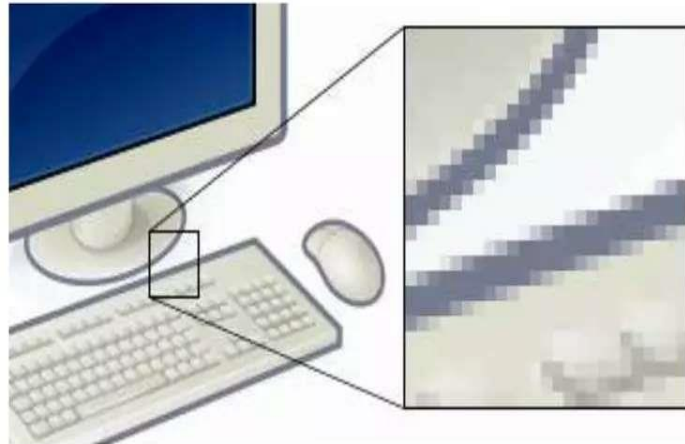


Topics:

- ❖ IMAGE RESOLUTION
 - ❖ DISPLAY RESOLUTION
 - ❖ MEGAPIXELS
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What is PIXEL?

- ❖ It is the smallest controllable element of a picture represented on the screen.



- ❖ This example shows an image with a portion greatly enlarged, in which the individual pixels are rendered as small squares and can easily be seen.

PIXEL need not to be square.

❖ A pixel does not need to be rendered as a small square.



❖ USING DOTS



USING LINES



USING SMOOTH FILTERING

❖ This image shows alternative ways of reconstructing an image from a set of pixel values, using dots, lines, or smooth filtering.

Pixel resolution

- ❖ Resolution is the capability of the sensor to observe or measure the smallest object clearly with distinct boundaries.
- ❖ There is a difference between the resolution and a pixel.
- ❖ A pixel is actually a unit of the digital image.
- ❖ Resolution depends upon the size of the pixel.
- ❖ With a given lens setting the smaller the size of the pixel, the higher the resolution will be and the clearer the object in the image will be.
- ❖ Images having smaller pixel sizes occupy more space on the disk.

Using pixels as a measure of resolution.

- ❖ The term "pixels" can be used in the abstract, or as a unit of measure.
- ❖ In particular when using pixels as a measure of resolution, such as: 2400 pixels per inch, 640 pixels per line, or spaced 10 pixels apart.....
- ❖ The measures dots per inch (dpi) and pixels per inch (ppi) are sometimes used interchangeably, but have distinct meanings, especially for printer devices, where;
- ❖ dpi is a measure of the printer's density of dot (e.g. ink droplet) placement. For example, a high-quality photographic image may be printed with 600 ppi on a 1200 dpi inkjet printer.
- ❖ Even higher dpi numbers, such as the 4800 dpi quoted by printer manufacturers since 2002, do not mean much in terms of achievable resolution.

PIXEL

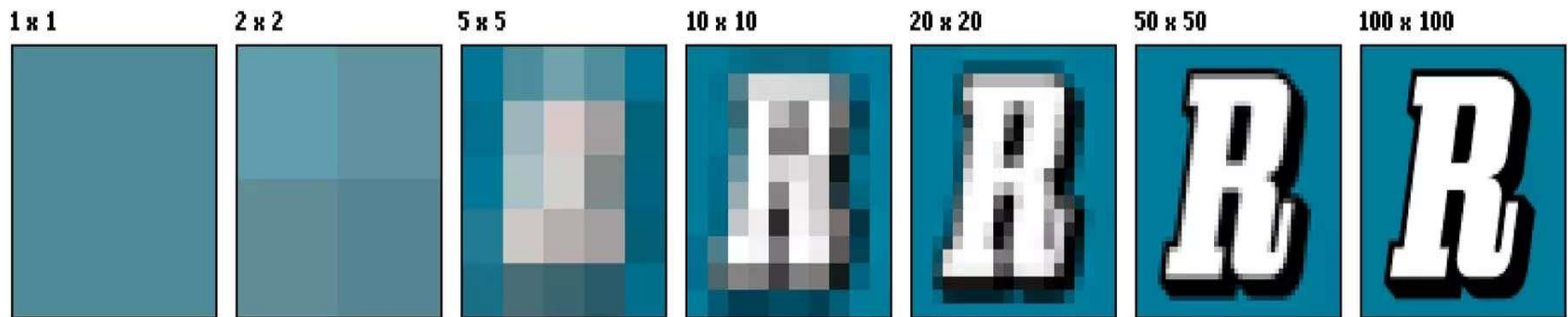
IMAGE
RESOLUTION

DISPLAY
RESOLUTION

Image resolution

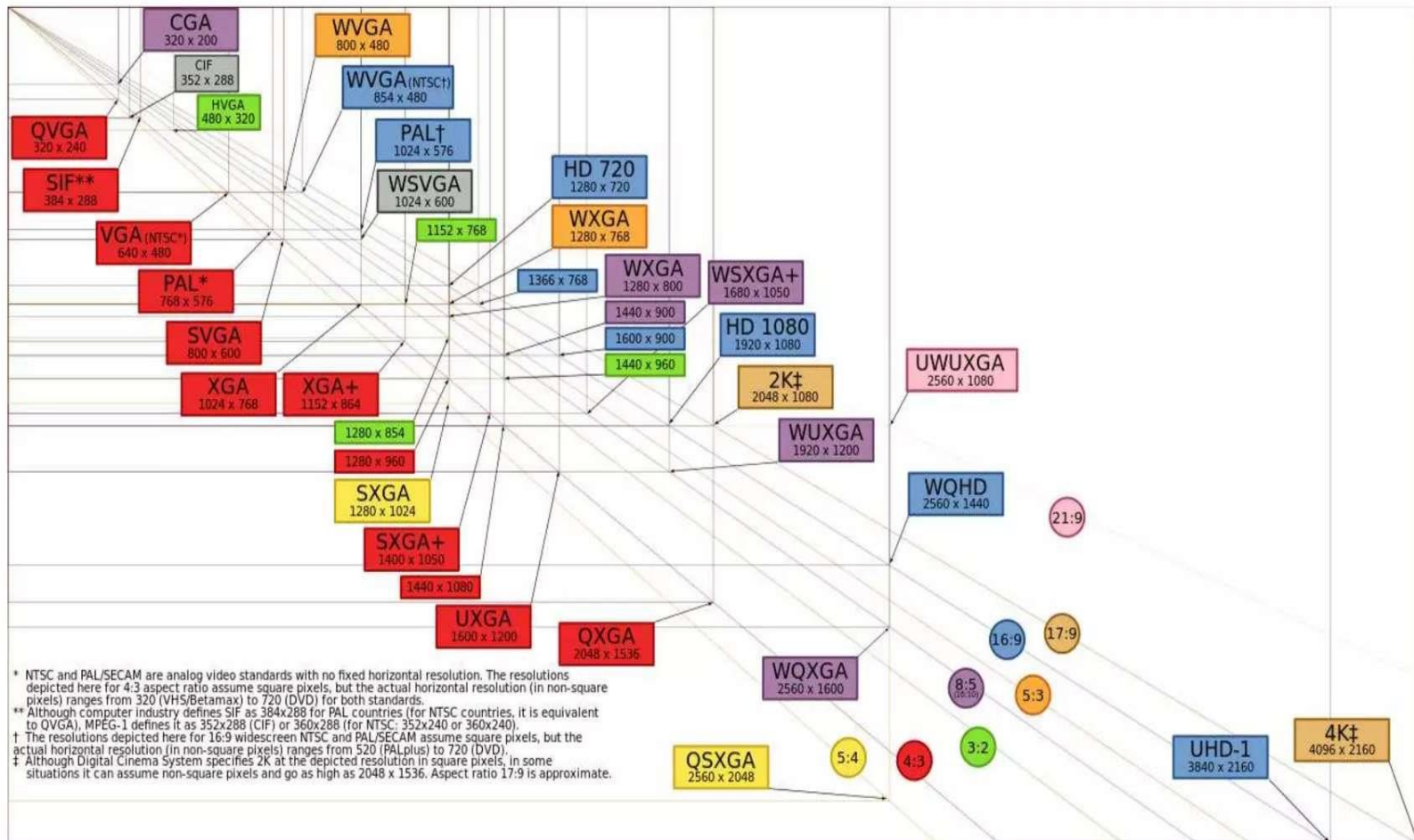
- ❖ **Image resolution** is the detail an image holds.
- ❖ Higher resolution means more image detail.
- ❖ Image resolution can be measured in various ways.
- ❖ Basically, resolution quantifies how close lines can be to each other and still be visibly resolved.
- ❖ A resolution 10 lines per millimeter means 5 dark lines alternating with 5 light lines, or 5 line pairs per millimeter (5 LP/mm)

Below is an illustration of how the same image might appear at different pixel resolutions, if the pixels were poorly rendered as sharp squares.



- ❖ An image that is 2048 pixels in width and 1536 pixels in height has a total of $2048 \times 1536 = 3,145,728$ pixels or 3.1 megapixels. One could refer to it as 2048 by 1536 or a 3.1-megapixel image.

Display resolution



Display resolution

- ❖ The **display resolution** of a Display resolution, computer monitor or display device is the number of distinct pixels in each dimension that can be displayed.
- ❖ It can be an ambiguous term especially as the displayed resolution is controlled by different factors in cathode ray tube(CRT), flat-panel display which includes liquid-crystal displays, or projection displays using fixed picture-element (pixel) arrays.
- ❖ It is usually quoted as **width × height**, with the units in pixels: for example, "1024 × 768" means the width is 1024 pixels and the height is 768 pixels.
- ❖ This example would normally be spoken as "ten twenty-four by seven sixty-eight" or "ten twenty-four by seven six eight".

Digital TV standards

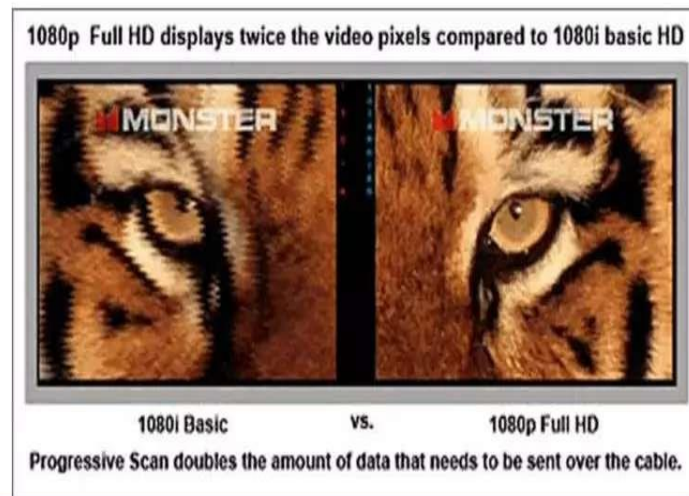
Standard	Resolution (dots × lines)	DAR (H:V)	Pixels
PixelVision	120 × 90	4:3	10,800
Video CD	352 × 240 (NTSC)	4:3	84,480
	352 × 288 (PAL)		101,376
UMD	480 × 272	~16:9	130,560
China Video Disc	352 × 480 (NTSC)	4:3 or 16:9	168,960
	352 × 576 (PAL)		202,725
SVCD	480 × 480 (NTSC)	4:3 or 16:9	230,400
	480 × 576 (PAL)		276,480
SDTV 480i , EDTV 480p , SMPTE 293M	640 × 480	4:3 or 16:9 or 3:2	307,200
	704 × 480		337,920
	720 × 480		345,600
	852 × 480		408,960
SDTV 576i , EDTV 576p	480 × 576	4:3 or 16:9	276,480
	544 × 576		313,344
	704 × 576		405,504
	720 × 576		414,720
	768 × 576		442,368
DVD	720 × 480 (NTSC)	4:3 or 16:9	345,600
	720 × 576 (PAL)		414,720
720p (HDTV, Blu-ray)	1280 × 720	16:9	921,600
1080i , 1080p (HDTV, Blu-ray)	1920 × 1080	16:9	2,073,600
2160p (UHDTV)	3840 × 2160	16:9	8,294,400
4320p (UHDTV)	7680 × 4320	16:9	33,177,600

What is the difference between SD, HD, XHD, and 4K?

- ❖ SD = Standard Definition = 480 interlaced lines (your old tube TV)
- ❖ ED = Enhanced Definition = 480 progressive lines (DVD Quality)
- ❖ HD = High Definition = could mean any of the following
 - ❖ 720p = 720 progressive lines (broadcast quality)
 - ❖ 1080i = 1080 interlaced (also broadcast quality)
 - ❖ 1080p = 1080 progressive lines (Bluray quality) highest quality for home theatre (double the resolution of 720p)
- ❖ 4K = 4K = high end movie theaters. If you saw District 9, or The Social Network, those were both filmed in 4k (instead of standard film). 4 times the resolution of 1080p. They've made 4K sets by combining 4 1080p screens.
- ❖ XHD = Extremely High Definition = 16x the resolution of 1080p. Largely theoretical.
You can't buy it for your house, or view it in the theatre.

Terminologies

- ❖ The "p" in 720p and 1080p refers to "**progressive**," which means a solid line of pixels across the screen.
- ❖ 1080i, on the other hand, means "**interlaced**," in which the odd pixels are shown on one scan of the screen and the even pixels on the next (the TV screen is scanned 60 times per second).
- ❖ The numbers themselves refer to the number of horizontal pixels the image resolution depicts, which is important when you want to differentiate between formats.



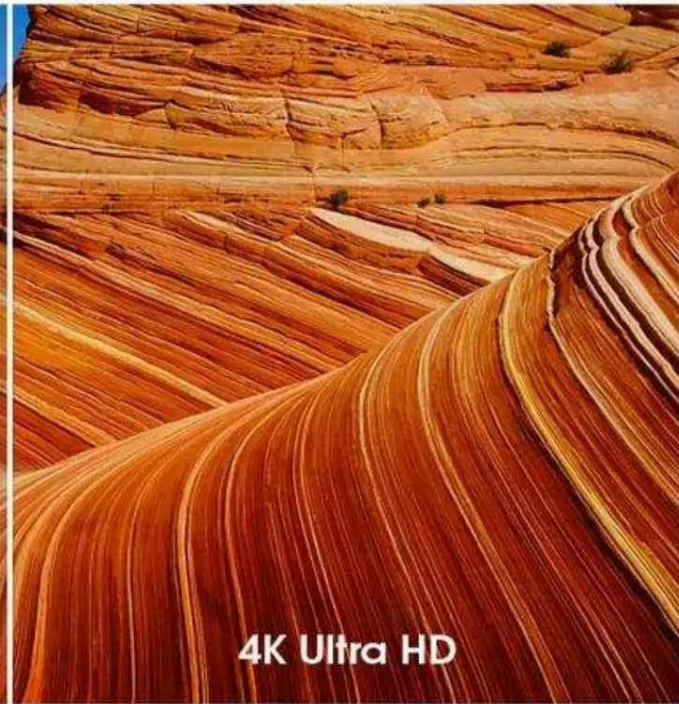
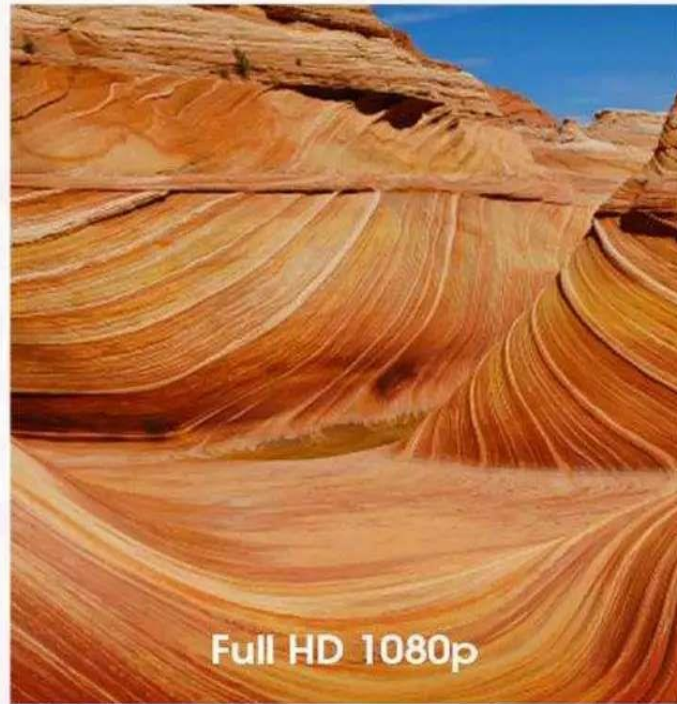
1080p

- ❖ The designation 1080p represents 1,080 vertical lines on your screen.
- ❖ The “p” stands for “progressive”, which means that all 1,080 vertical lines are shown at the same time.
- ❖ When you add in the horizontal dimensions of 16:9 the total pixels number around **two million**.
- ❖ The quality of the visuals is divided equally between our two eyes.
- ❖ This means that although you do see a higher-quality picture than was once available on the old cathode ray tube sets of yesteryear, your mind still only sees half of the resolution per eye.
- ❖ This significantly reduces the clarity of what you are seeing.

4k

- ❖ With the new Ultra High-Definition 4K televisions the lines are counted horizontally at **4,320** lines.
- ❖ The total pixels on such a television are just **over eight million**.
- ❖ This means that each eye is able to see a full high-definition image, essentially if each eye were individually tuned to a 1080p screen.
- ❖ The quality far exceeds that we have in our old high def TVs, and makes advancements such as three dimensional imagery possible.
- ❖ This means that essentially 4k UHD TVs have **4 times more detail than Full HD 1080p**.

1080p vs 4k



Comparision

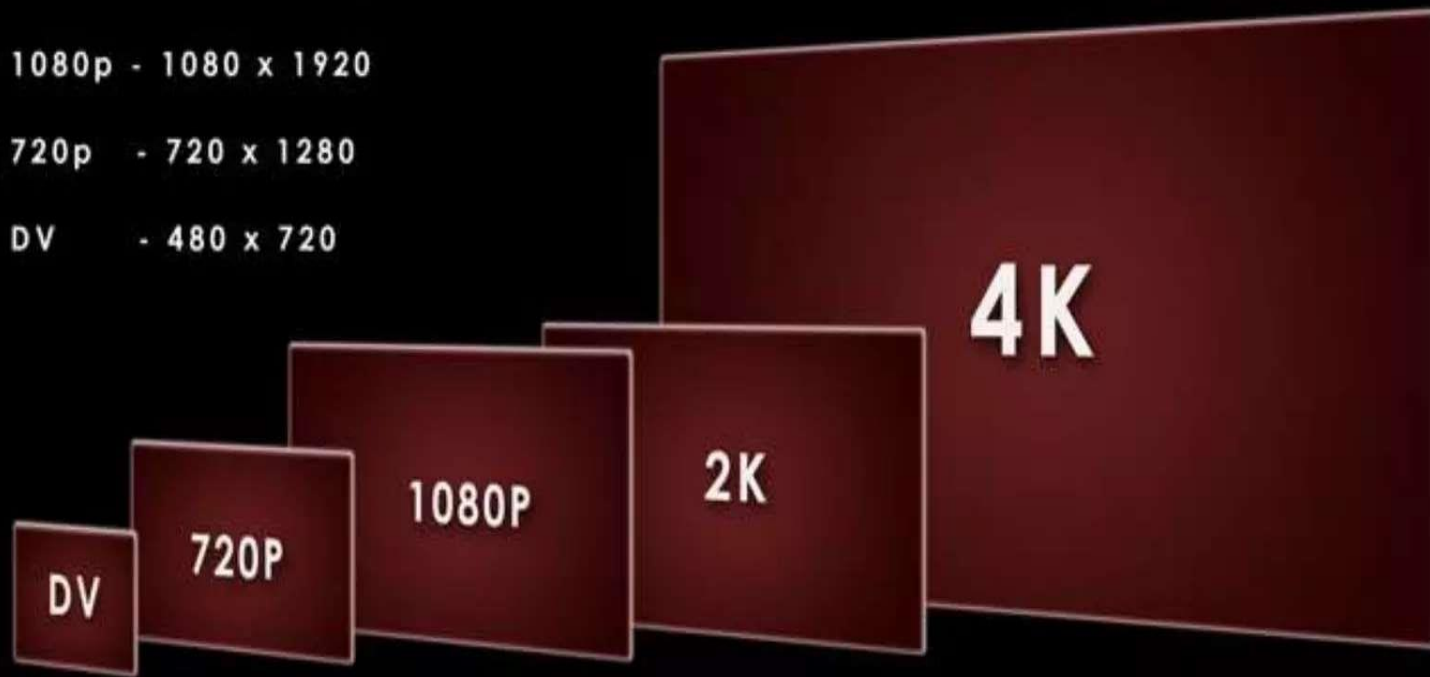
4K - 2304 x 4096

2K - 1152 x 2048

1080p - 1080 x 1920

720p - 720 x 1280

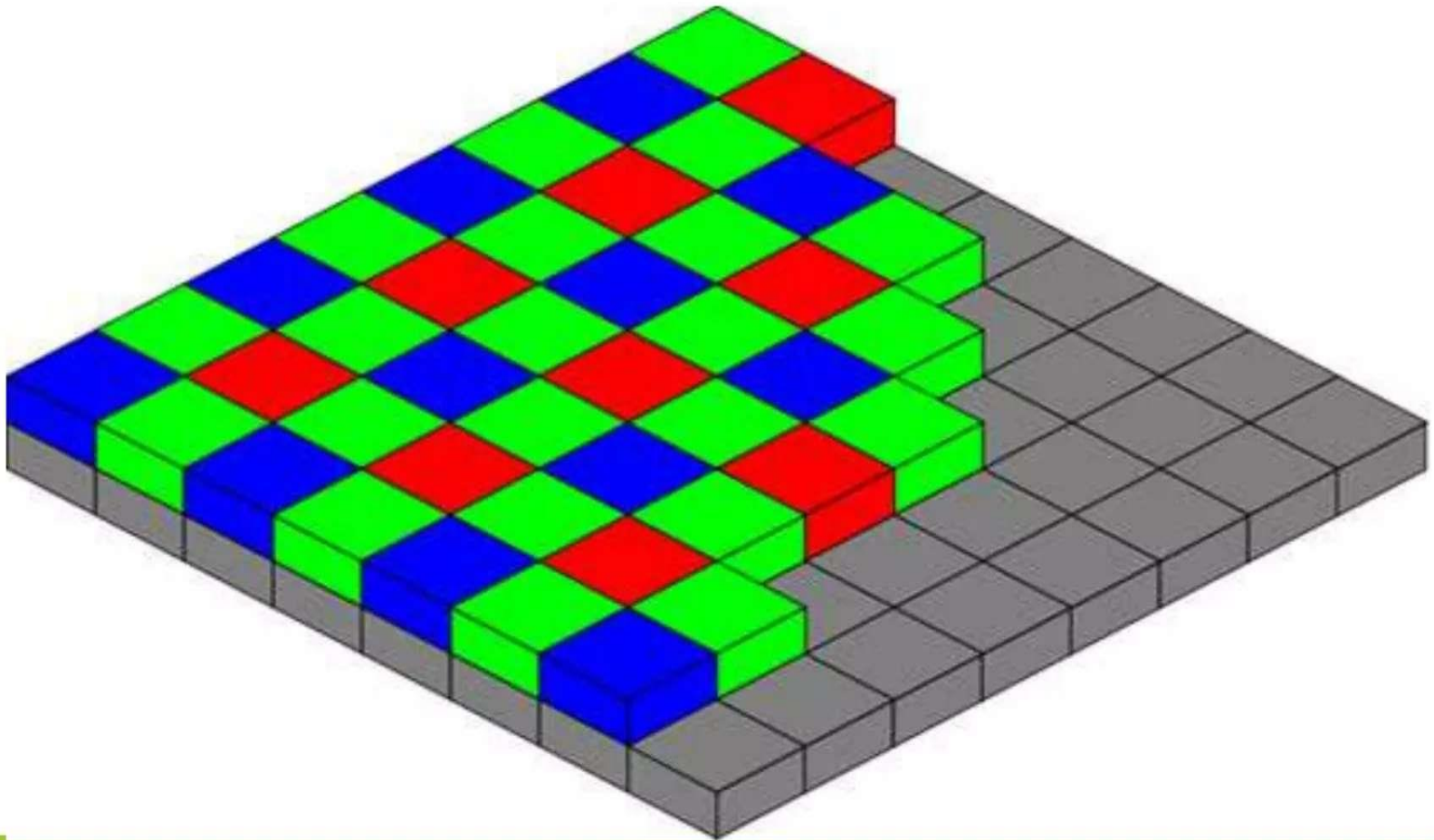
DV - 480 x 720



Facts

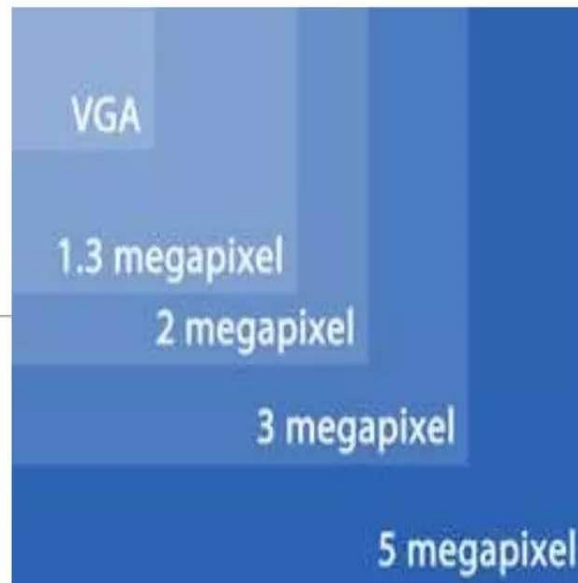
- ❖ Well 4K in the world of the professionals who do this, and you say "4K," it means you have 4096 red, 4096 green and 4096 blue photo sites.
- ❖ In other words...
- ❖ In order to get RGB out of a Bayer pattern you need two lines.
- ❖ Because you only have green plus one color (red) on one line, and green plus the other color (blue) on the other line.
- ❖ You then have to interpolate the colors that are missing from surrounding pixels.

To create a single RGB pixel, there must be an equal number of each color



Megapixel

- ❖ This term refers to the size of an image, usually in reference to a photo from a digital camera or camera phone.
- ❖ Megapixel means one million pixels. The resolution of digital cameras and camera phones is often measured in megapixels. For example, a 5-megapixel camera can produce images with five million total pixels.
- ❖ Since pixels are usually square and form a grid, a 1-megapixel camera will produce an image roughly 1200 pixels wide by 900 pixels high.
- ❖ Rather, high-megapixel cameras are most useful for taking photos suitable for viewing on a PC and/or printing.
- ❖ However, more megapixels do matter when viewed on a phone, when zooming in or cropping a photo



- ❖ For example, some phones with 20 or more megapixels let users "zoom in" without losing quality, without an optical zoom lens. They do this by simply cropping a high-quality 8-megapixel photo from the center of the 41-megapixel image captured by the camera (for example.)
- ❖ Photos with more megapixels are larger in file size, and therefore can take longer - and cost more - to send over the network at full size.
- ❖ Most cameras and camera phones have an option to take photos at lower resolution (smaller), if desired.
- ❖ This is useful for taking photos that will only be sent via MMS and not transferred to a PC or printed.